

MOUNTING HOLES

MAIN BOARD

- H1 MountingHole
- H2 MountingHole
- H3 MountingHole
- H4 MountingHole

GNSS BOARD

- H5 MountingHole
- H6 MountingHole
- H7 MountingHole
- H8 MountingHole

GNSS BOARD

- | | |
|--|--|
|  H1 |  H4 |
|  MountingHole |  MountingHole |
|  H2 |  H5 |
|  MountingHole |  MountingHole |
|  H3 |  H6 |
|  MountingHole |  MountingHole |

DATA STORAGE

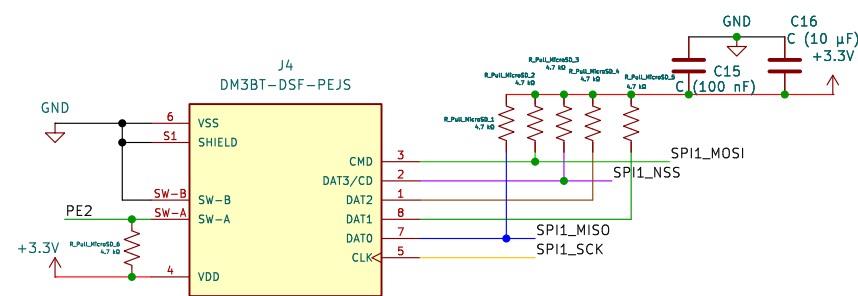
DM3BT-DSF-PEJS – MICROSD CONNECTOR

The diagram illustrates the electrical connections for the DM3BT-DSF-PEJS MicroSD connector. The connector is represented as a yellow block with pins 1 through 9. The internal components and connections are as follows:

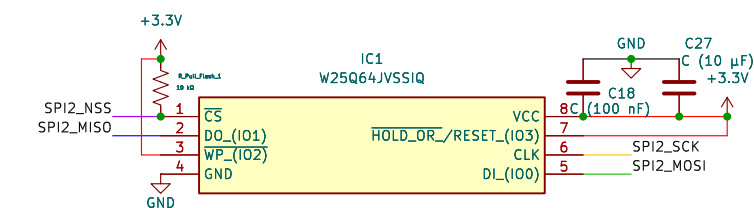
- Power and Ground:**
 - VSS (Pin 6):** Connected to GND.
 - VDD (Pin 4):** Connected to +3.3V through a 10kΩ pull-up resistor.
 - SW-A (Pin 5):** Connected to +3.3V through a 10kΩ pull-up resistor.
 - SW-B (Pin 3):** Connected to +3.3V through a 10kΩ pull-up resistor.
- Signal Lines:**
 - CMD (Pin 3):** Connected to SPI1_NSS.
 - DAT3/CD (Pin 2):** Connected to SPI1_MISO.
 - DAT2 (Pin 1):** Connected to SPI1_MOSI.
 - DAT1 (Pin 8):** Connected to SPI1_MISO.
 - DAT0 (Pin 7):** Connected to SPI1_MOSI.
 - CLK (Pin 9):** Connected to SPI1_SCK.
- Other Connections:**
 - PE2 (Pin 2):** Connected to GND.
 - SW-B (Pin 3):** Connected to GND.
 - SW-A (Pin 5):** Connected to GND.

The circuit also includes several capacitors: C15 (100 nF) and C16 (10 μF) connected to GND, and C17 (10 nF) connected to the SPI1_NSS line. The connector is labeled J4 and DM3BT-DSF-PEJS.

DM3BT-DSF-PEJS – MICROSD CONNECTOR



W25Q64 – FLASH CHIP

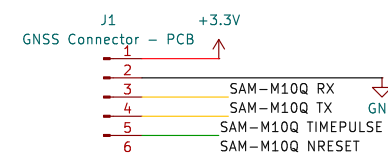


GNSS

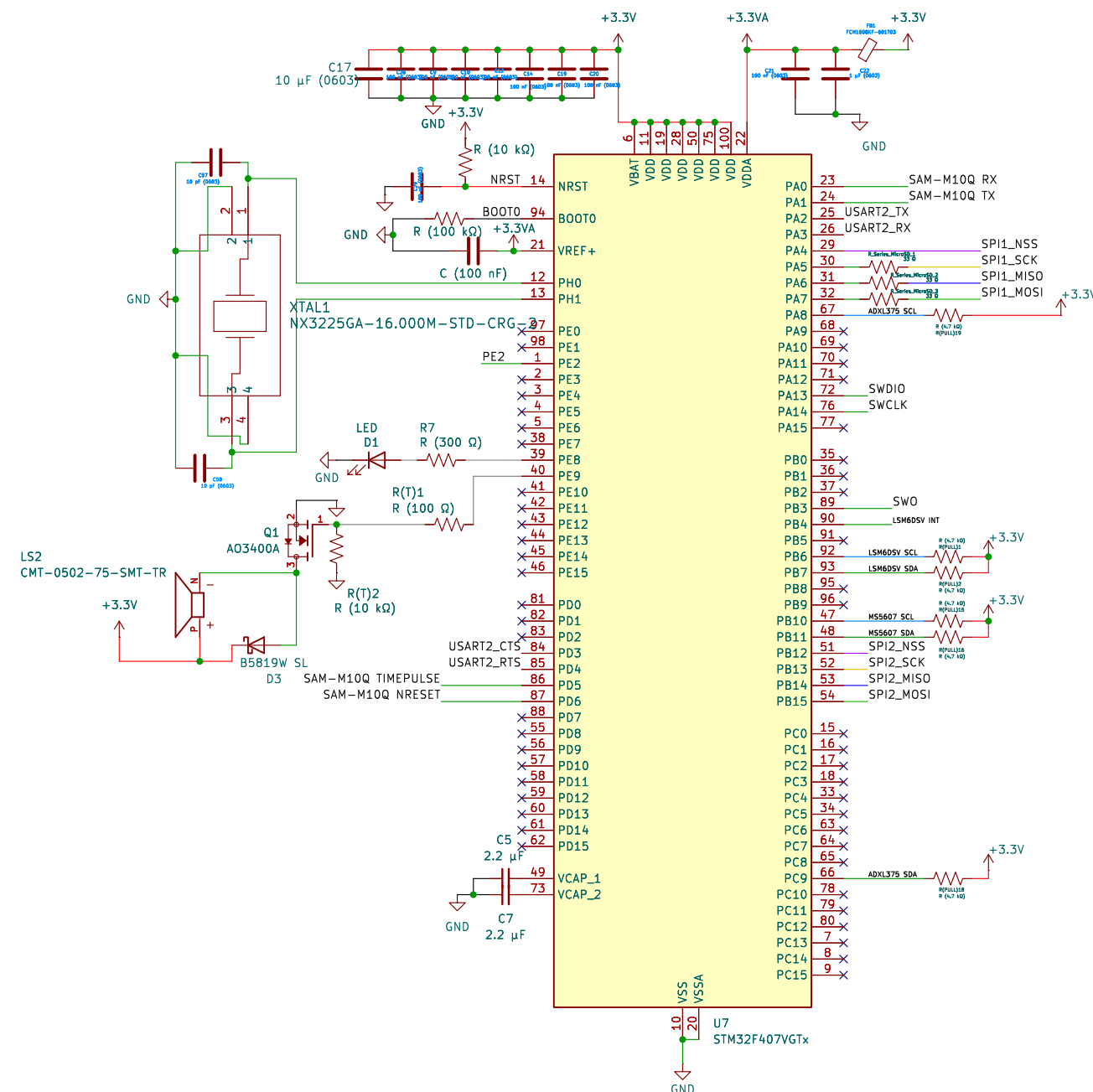
SAM-M10Q – GNSS MODULE

SAM-M10Q – GNSS MODULE

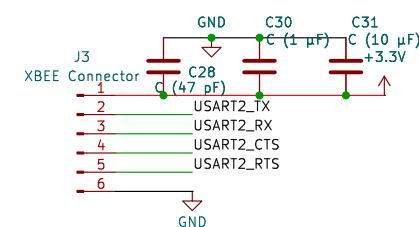
JST-GH GNSS CONNECTORS



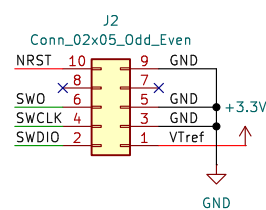
STM32F407 – MICROCONTROLLER



XBEE CONNECTOR



JTAG/SWD PINS



BUCK CONVERTER

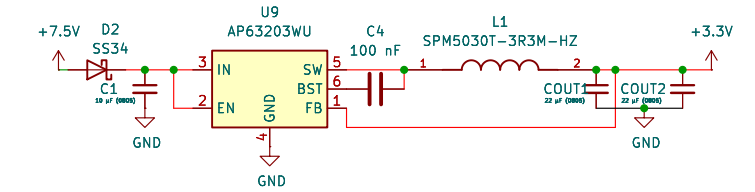
AP63203 – BUCK SWITCHING REGULATOR

The diagram shows a BUCK converter circuit. The input is +7.5V connected to pin 3 (IN) of the AP63203WU. Pin 2 (EN) is connected to GND. Pin 5 (SW) is connected to a 100 nF capacitor (C4) and the positive terminal of an inductor (L1, SPM5030T-3R3M-HZ). The other end of the inductor is connected to pin 6 (BST) and the positive terminal of an output capacitor (COUT2, 22 μF). Pin 4 (GND) and pin 1 (FB) are connected to GND. The output of the converter is +3.3V, connected to pin 7 (GND) and the negative terminal of the output capacitor. A diode (D2, SS34) is connected in parallel with the output capacitor, with its cathode to the +3.3V line and its anode to GND.

XT30 CONNECTORS

The diagram shows two XT30 connectors. The first connector, labeled J5 Conn_01x02, has a +7.5V input connected to pin 2 and GND connected to pin 1. The second connector, labeled Power_On_LED1 RED, has a +3.3V input connected to pin 2 and a red LED connected to pin 1. The LED is connected in series with a resistor (R, 330 Ω) to GND.

AP63203 – BUCK SWITCHING REGULATOR



XT30 CONNECTORS



SENSOR SUITE

MS5607 – BAROMETRIC PRESSURE SENSOR

U5
MS560702BA03-50

+3.3V
GND

MS5607_SCL
MS5607_SDA

CSB
PS
SCLK
SDO/SDA

VDD
GND

C11 (100 nF)

ADXL375 – HIGH G ACCELEROMETER

U4
ADXL375BCCZ

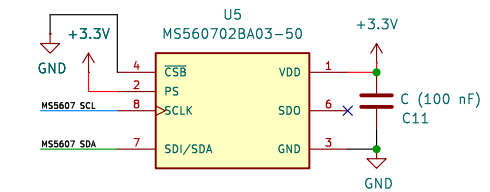
+3.3V
GND

ADXL375_SCL
ADXL375_SDA

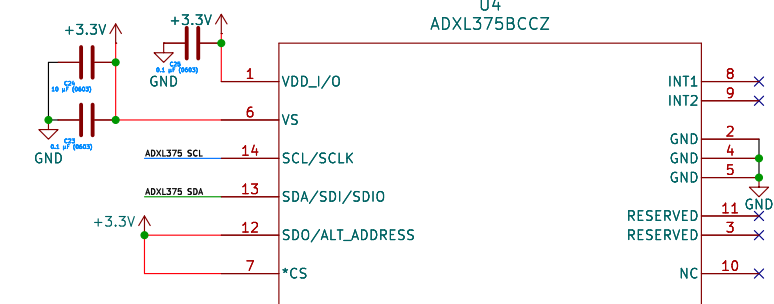
VDD_I/O
VS
SCL/SCLK
SDA/SDI/SDIO
SDO/ALT_ADDRESS
*CS

INT1
INT2
GND
GND
GND
RESERVED
RESERVED
NC

MS5607 – BAROMETRIC PRESSURE SENSOR



ADXL375 – HIGH G ACCELEROMETER



LSM6DSV32 – 6-AXIS INERTIAL MEASUREMENT UNIT

